

Healthcare Appointment No-Show Prediction Report

Abstract

This project aims to predict whether a patient will miss a medical appointment using machine learning techniques. A Decision Tree model is trained on historical appointment data to identify patterns associated with no-shows. Key factors such as waiting time, age, SMS reminders, and appointment day are analyzed. The results help hospitals identify high-risk patients and optimize scheduling, reduce resource wastage, and improve operational efficiency.

Introduction

Missed medical appointments lead to inefficient use of healthcare resources, increased waiting times, and financial losses. With the help of data analytics and machine learning, hospitals can predict patient attendance behavior in advance. This project focuses on analyzing appointment data to understand no-show patterns and build a predictive model that supports better scheduling decisions and patient management.

Tools Used

- Python (Pandas, NumPy) for data processing
- Scikit-learn for Decision Tree model
- Matplotlib for feature importance visualization
- VS Code for development
- Tableau for interactive dashboard and insights

Steps Involved in Building the Project

- Import and clean the healthcare appointment dataset
- Perform feature engineering such as Waiting Days and Weekday
- Encode categorical variables for model training
- Split data into training and testing sets
- Train a Decision Tree model to predict no-shows
- Evaluate model performance using accuracy and confusion matrix
- Export cleaned data and predictions for Tableau
- Create interactive dashboard to analyze key trends

Conclusion

The project successfully predicts patient no-show behavior and identifies major influencing factors such as waiting time, age, and SMS reminders. The insights generated through Tableau help hospitals take proactive actions such as sending reminders, reducing waiting periods, and applying controlled overbooking. Implementing such predictive systems can improve resource utilization, reduce revenue loss, and enhance overall healthcare service efficiency.